

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (CAD/CAM)

Sem - I University Examination, 2009-10

ME – 701 Computer Aided Design

Date: 26.12.2009, Saturday

Time: 01:00 p.m. to 04:00 p.m.

Maximum Marks: 70

## Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.
3. Use of PSG Design data book is permitted.

## SECTION- I

- Q1 (a) Explain the following solid modelling schemes with suitable examples; 04
- 1) Constructive Solid Geometry
  - 2) Boundary Representation
- (b) Differentiate between; 04
- 1) Surface modelling & Solid modelling
  - 2) Ruled surface & Tabulated surface
- (c) State the modelling facilities that are expected in good solid modelling software. 05
- Q2 (a) With the help of syntax, state the use of the following graphics functions in C/C++; 02
- 1) initgraph ( )
  - 2) detectgraph ( )
- (b) Explain: "Bresenham's algorithm is more efficient than the DDA line generation algorithm." 02
- (c) Why can't a vector scan display produce as many colours as a raster scan display? 02
- (d) What is scan conversion? Write an algorithm to scan convert a line having slope more than  $45^\circ$  using DDA algorithm. 05

OR

- Q2 (a) Differentiate between: 02
- 1) Printer & Plotter
  - 2) Mouse and Digitizer
- (b) Explain the working principle of raster scan display devices with the help of neat sketches. 04
- (c) Write Bresenham's circle drawing algorithm with  $(x_c, y_c)$  as center and radius  $r$ . 05
- Q3 (a) Explain non-parametric form of representing curves. What are the limitations of this form of representation for CAD/CAM systems? 03
- (b) Show that the transformation matrix for reflection about the line  $Y = -X$  is equivalent to reflection relative to Y axis followed by counterclockwise rotation by  $90^\circ$ . 04
- (c) Explain homogeneous coordinate system, & differentiate between geometric transformation & coordinates transformation. 04

OR

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- Q3 (a)** Explain the curves having following conditions: 04
- 1) Zero order continuity
  - 2) First order continuity
  - 3) Second order continuity
- (b)** Distinguish between Model Coordinate System, World Coordinate System & Screen Coordinate System. 04
- (c)** Derive from fundamentals the necessary transformation matrix for rotating an object in anticlockwise direction by an angle  $\theta$  about Y-axis. 04

## SECTION - II

- Q4 (a)** Explain the following terms with respect to clipping: 04
- 1) World Coordinate System
  - 2) Window
  - 3) Image Coordinate System
  - 4) Viewport
- (b)** Give the steps for building up a computerized geometric solid model of a "Flange" using any solid modeling software. 04
- (c)** Describe the concept of parametric programming. Write a parametric program to draw a locating pin. 04
- Q5 (a)** Explain in brief different type of scanners. What are active scanner and passive scanner? 05
- (b)** The coordinates of four control points relative to a current WCS are given by  $P_0 = [2 \ 2 \ 0]^T$ ,  $P_1 = [2 \ 3 \ 0]^T$ ,  $P_2 = [3 \ 3 \ 0]^T$  &  $P_3 = [3 \ 2 \ 0]^T$ . Find the equation of the resulting Bezier curve. Also find points on curve for  $u = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \text{ \& } 1$ . 06

OR

- Q5 (a)** Explain rotary mode and plane mode of scanning techniques. 03
- (b)** List the various properties of Bezier curves 04
- (c)** Calculate the parametric mid point of the Hermite cubic curve that fits the points  $P_0 (1, 1)$ ,  $P_1 (6, 5)$  & the tangent vectors  $P_0' (0, 4)$ ,  $P_1' (4, 0)$ . 04
- Q6 (a)** Write a short notes on the following: 06
- 1) DXF - Format
  - 2) STL - Format
- (b)** Develop an algorithm & interactive computer program for design of connecting rod of an IC engine. 06

OR

- Q6 (a)** Write a short notes on the following: 06
- 1) GKS
  - 2) NURBS
- (b)** For design of helical compression springs prepare an algorithm & interactive programme using any programming language. 06